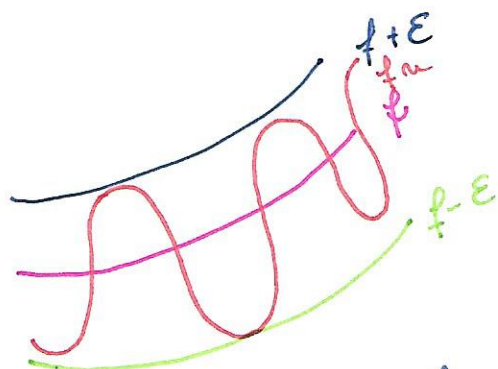




$$f_n(x) = x^n \rightarrow \begin{cases} 0, & x \in [0, 1) \\ 1, & x = 1 \end{cases}$$

is an example of non-uniform convergence

T3. If the functions $f_n : [a, b] \rightarrow \mathbb{R}$ are continuous and if $f_n \xrightarrow{u} f$, then f is continuous on the closed interval $[a, b]$.

T4 Let (f_n) be a uniformly convergent sequence of continuous functions on the interval $[a, b]$. Then the following equality holds

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx$$

By virtue of the previous theorem we deduce

C5 A uniformly convergent series of continuous functions on a closed interval $[a, b]$ can be integrated termwise (term-by-term)

T6. Let (f_n) be a sequence of functions in $C^1[a, b]$. If (f_n) converges to a function f , and the sequence of derivatives (f_n') is uniformly convergent to a function g , then there exists f' and $f' = g$, i.e.

$$\lim_{n \rightarrow \infty} (f_n)' = (\lim_{n \rightarrow \infty} f_n)'$$

T7 (Weierstrass) If $\sum a_n$ is convergent series of numbers, and the functional sequence of real functions (f_n) , satisfies the inequalities

$$|f_n(x)| \leq a_n \quad (\forall x \in A; \forall n \in \mathbb{N})$$

then the functional series $\sum f_n$ is uniformly and absolutely convergent on A .

Power Series

- puten serie in calculator once function.

Let (a_n) be a sequence of complex numbers and $z_0 \in \mathbb{C}$

D8 The series

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

is called a power series in powers of $z - z_0$ and coefficients (a_n) .